

Isogeny Computations in the Level-4 Mumford Model

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Motivation & Context

The Isogeny Landscape

Efficient isogeny evaluation is central in various number-theoretic applications, including isogeny-based post-quantum cryptography.

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Currently, algorithmic efficiency is tied to specific curve models, like:

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- Twisted Edwards/Hesse curves.

However, relying on a narrow set of curve families limits our algebraic flexibility and design space.

The Challenge

- **Performance Bottleneck:** Computing isogenies and scalar multiplications can require *heavy* finite field arithmetic.
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- **Core Metric:** We measure success by minimizing the number of field multiplications (**M**) and squarings (**S**).

To optimize these counts, we don't just optimize algorithms - we look for alternative geometric models of elliptic curves that possess favorable arithmetic properties.

The description of N -isogeny formulas becomes particularly simple when working with a level- N model.

- SoTA mainly focuses on level-2 and -3 models.

The Study Case: Level-4

Research Question: can we find a model that supports level-4 structures *efficiently* provides an alternative mathematical foundation?

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Aim of Our Work: determine whether the new interpretation of degree-4 isogeny formulas leads to improvements of the arithmetic in this model.

Elliptic curves in level-4 Mumford model
and the Kummer quotient(s)

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Radical 4-isogenies and multiplication-by-4

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The Level-4 Mumford Model & The Quotients

How a Theta Structure Induces a Model

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The mechanism operates in various steps, including:

- identifying the projective embedding

$$P \mapsto [\theta_0(P) : \theta_1(P) : \cdots : \theta_{N-1}(P)] \in \mathbb{P}^{N-1};$$

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The coefficients of the resulting model are entirely determined by evaluating the theta functions at the neutral element.

From the Mumford-4 Model to the Kummer Line

As a direct consequence of this machinery, a level-4 theta structure induces a model of an elliptic curve E (over $\text{char}(p) \neq 2$) in $\mathbb{P}^3$ as the intersection of two quadrics:

$$E = E_{(a:b)} : \begin{cases} b(x_1^2 + x_3^2) = 2ax_0x_2 \\ b(x_0^2 + x_2^2) = 2ax_1x_3 \end{cases}$$

for some projective coefficient

$$(a : b) \notin \{(1 : \pm 1), (1 : \pm i), (1 : 0), (0 : 1)\}.$$

From the Mumford-4 Model to the Kummer Line

Computing directly on the full curve can be slow and inefficient, so we project down to:

- the Kummer line $E/\langle \pm 1 \rangle$,
- a degree-2 quotient of the Kummer line,

in order to successfully derive formulas for $4 \cdot P$ bypassing the full curve arithmetic.

From the Mumford-4 Model to the Kummer Line

Multiplication by $[-1]$ on E in Mumford form acts by a permutation of the coordinates:

$$\begin{aligned} [-1] : E &\longmapsto E \\ (x_0 : x_1 : x_2 : x_3) &\longmapsto (x_0 : x_3 : x_2 : x_1). \end{aligned}$$

The *Kummer* coordinates $u_0 = x_0$, $u_1 = x_1 + x_3$, $u_2 = x_2$ are invariant by multiplication with $[-1]$.

They are related by the quadratic *Kummer* equation

$$\mathcal{K} : abu_1^2 = 2a^2u_0u_2 + b^2(u_0^2 + u_2^2).$$

From the Kummer line to a special quotient [name tbd]

In addition to this model of the Kummer line, we are further interested in the *special* quotient of E

$$\overline{\mathcal{K}} = \mathcal{K} / \langle \pm u_1 \rangle \cong \mathbb{P}^1.$$

The doubling formulas for E descend to this quotient as well, and are remarkably similar to doubling formulas for level-2 theta functions.

4-isogenies

The 4-torsion and symplectic transformations

If $0_E = (t_0 : t_1 : t_2 : t_1)$ is the neutral element of E , then $E[4] = \langle P, Q \rangle$ with

$$P = (t_1 : t_2 : t_1 : t_0), \quad Q = (t_0 : it_1 : -t_2 : -it_1)$$

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Having an explicit description of the 4-torsion points allows us to:

- describe the action of P, Q on a generic point $X \in E$;
- characterize certain 8-torsion points;
- provide symplectic transformations of $\mathbb{P}^3$ acting on this 4-torsion basis.

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Symplectic transformations are linear transformations that define isomorphisms of elliptic curves in Mumford form; specifically, fixed a primitive 8-th root of unity ζ , the matrices

$$D = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -i & -1 & i \\ 1 & -1 & 1 & -1 \\ 1 & i & -1 & -i \end{bmatrix}, \quad M = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \zeta & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & \zeta \end{bmatrix},$$

viewed as linear transformations of $\mathbb{P}^3$, define those isomorphisms:

$$\begin{aligned} f_D : E_{(a:b)} &\rightarrow E_{(a+b:a-b)}, \\ x &\mapsto D \cdot x \end{aligned}$$

$$\begin{aligned} f_M : E_{(a:b)} &\rightarrow E_{(ia:b)}, \\ x &\mapsto M \cdot x. \end{aligned}$$

The 4-torsion and symplectic transformations

Moreover, the action on the canonical 4-torsion bases is given by

$$F_D = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, \quad F_M = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}.$$

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These transformations will be useful when we want to transform the kernel of a 4-isogeny into a specific form:

- there are 6 outgoing 4-isogenies at $E_{(a:b)}$ with kernels

$$\langle P \rangle, \quad \langle P + Q \rangle, \quad \langle P + 2Q \rangle, \quad \langle P + 3Q \rangle, \quad \langle Q \rangle, \quad \langle Q + 2P \rangle;$$

- They are respectively transformed via the following transformations to a canonical form:

$$D, \quad DM, \quad DM^2, \quad DM^3, \quad Id, \quad MD^{-2}M^{-1}.$$

4-isogenies

In order to describe 4-isogenies, we introduce the following:

- $\mathcal{F}$: discrete Fourier transform;
- $\mathcal{C}_\lambda$ with $\lambda = (\lambda_0 : \lambda_1 : \lambda_2) \in \mathbb{P}^2$: coordinate-wise scaling
 $\mathcal{C}_\lambda(x) = (\lambda_0 x_0 : \lambda_1 x_1 : \lambda_2 x_2 : \lambda_1 x_3);$
- $\odot^4$: coordinate-wise fourth powers
 $\odot^4(x) \mapsto (x_0^4 : x_1^4 : x_2^4 : x_3^4).$

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If $R \in E[32]$ is s.t. $8 \cdot R = Q$ and $(r_0 : r_1 : r_2 : r_3) = \mathcal{F} \circ \odot^4(R)$, then

$$\phi = \mathcal{F} \circ \mathcal{C}_\lambda \circ \mathcal{F} \circ \odot^4 : E \rightarrow E'$$

with $\lambda = (r_1 r_2 : r_0 r_2 : r_0 r_3)$ defines a 4-isogeny with kernel $\langle Q \rangle$. Moreover, the codomain E' is in level-4 Mumford form, and the canonical 4-torsion basis is given by $(\phi(P), 2 \cdot \phi(R))$.

4-isogenies on the Kummer line

The maps $\odot^4, \mathcal{F}, \mathcal{C}_\lambda$ descend on the Kummer line as follows:

$$\begin{pmatrix} u_0 \\ u_1 \\ u_2 \end{pmatrix} \xrightarrow{\widetilde{\odot^4}} \begin{pmatrix} 2a^2b^2u_0^4 \\ 8a^4u_0^2u_2^2 - b^4(u_0^2 + u_2^2)^2 \\ 2a^2b^2u_2^4 \end{pmatrix},$$

$$\begin{pmatrix} u'_0 \\ u'_1 \\ u'_2 \end{pmatrix} \xrightarrow{\widetilde{\mathcal{F}}} \begin{pmatrix} u'_0 + u'_1 + u'_2 \\ 2(u'_0 - u'_2) \\ u'_0 - u'_1 + u'_2 \end{pmatrix}, \quad \begin{pmatrix} u''_0 \\ u''_1 \\ u''_2 \end{pmatrix} \xrightarrow{\widetilde{\mathcal{C}_\lambda}} \begin{pmatrix} \lambda_0 u''_0 \\ \lambda_1 u''_1 \\ \lambda_2 u''_2 \end{pmatrix}.$$

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$$\begin{pmatrix} u'_0 \\ u'_1 \\ u'_2 \end{pmatrix} \xrightarrow{\widetilde{\mathcal{F}}} \begin{pmatrix} u'_0 + u'_1 + u'_2 \\ 2(u'_0 - u'_2) \\ u'_0 - u'_1 + u'_2 \end{pmatrix}, \quad \begin{pmatrix} u''_0 \\ u''_1 \\ u''_2 \end{pmatrix} \xrightarrow{\widetilde{\mathcal{C}_\lambda}} \begin{pmatrix} \lambda_0 u''_0 \\ \lambda_1 u''_1 \\ \lambda_2 u''_2 \end{pmatrix}.$$

The 4-isogeny $\widetilde{\phi} : \mathcal{K} \rightarrow \mathcal{K}'$ is given by $\widetilde{\mathcal{F}} \circ \widetilde{\mathcal{C}_\lambda} \circ \widetilde{\mathcal{F}} \circ \widetilde{\odot^4}$, and the evaluation of a 4-isogeny on the Kummer line costs $4S + 16a + 7m$.

4-isogenies from the special quotient (work in progress)

In analogous way to the Kummer line, we find an isogeny on the *special* quotient that lifts to the 4-isogeny ϕ , and is explicitly defined through the use of 16-torsion and the following maps $\mathbb{P}^1 \rightarrow \mathbb{P}^1$:

- $\mathcal{H}$: Hadamard transform;
- $\mathcal{C}_\lambda$ with $\lambda = (\lambda_0 : \lambda_2) \in \mathbb{P}^1$: coordinate-wise scaling
 $\mathcal{C}_\lambda(x) = (\lambda_0 x_0 : \lambda_2 x_2)$;
- $\odot^2$: coordinate-wise squaring $\odot^2(x) \mapsto (x_0^2 : x_2^2)$.

Radical 4-isogenies and multiplication-by-4

Radical isogenies

A *radical N -isogeny formula* takes as input an elliptic curve E together with a point P of order N . The output consists of:

- an elliptic curve E' which is the codomain of an N -isogeny $\phi : E \rightarrow E'$ with kernel $\ker(\phi) = \langle P \rangle$;
- a point $P' \in E'[N]$ which is a point of order N defining an isogeny $\phi' : E' \rightarrow E'/\langle P' \rangle$ with the property that $\phi' \circ \phi$ is an N^2 -isogeny.

The formula itself consists of algebraic expressions in the coefficients of E and the coordinates of P , as well as one N -th root.

Note: the neutral element $(t_0 : t_1 : t_2 : t_1)$ already encodes all information about the elliptic curve, and a 4-torsion basis, so we only have to provide:

- the coordinates of the neutral element on the domain,
- the codomain of a 4-isogeny.

Radical 4-isogenies (theorem)

Let E be an elliptic curve in level-4 Mumford form with neutral element $0_E = (t_0 : t_1 : t_2 : t_1)$. And consider the elliptic curve E' with

$$0_{E'} = (t_0 + \alpha : t_2 : t_0 - \alpha : t_2), \quad \text{with } \alpha = \sqrt[4]{t_0^4 - t_2^4}.$$

Further, let (P, Q) and (P', Q') be the canonical bases of $E[4]$ and $E'[4]$, respectively.

Then there exists a 4-isogeny with $\phi : E \rightarrow E'$ with kernel $\ker(\phi) = \langle Q \rangle$, and the kernel of the dual isogeny is given by $\ker(\hat{\phi}) = \langle P' \rangle$.

In particular, the above is a *radical isogeny formula*.

Multiplication-by-4 (theorem)

Let E be an elliptic curve in level-4 Mumford form with neutral element $0_E = (t_0 : t_1 : t_2 : t_1)$. Set $\lambda = (\lambda_0 : \cdots : \lambda_3)$ and $\lambda' = (\lambda'_0 : \cdots : \lambda'_3)$ with

$$\begin{cases} \lambda_0 := (t_0^3 - t_2^3)^4(t_0^4 - t_2^4)^3, \\ \lambda_1 := (t_0^6 - t_2^6)^4, \\ \lambda_2 := (t_0^3 + t_2^3)^4(t_0^4 - t_2^4)^3, \\ \lambda_3 := (t_0^6 - t_2^6)^4, \end{cases} \quad \begin{cases} \lambda'_0 := 4t_2t_1^3(3t_0^2 + t_2^2), \\ \lambda'_1 := t_0t_2(3t_0^2 + t_2^2)(t_0^2 + 3t_2^2), \\ \lambda'_2 := 4t_0t_1^3(t_0^2 + 3t_2^2), \\ \lambda'_3 := t_0t_2(3t_0^2 + t_2^2)(t_0^2 + 3t_2^2). \end{cases}$$

Then the multiplication-by-4 map $[4] : E \rightarrow E$ can be decomposed into simple operations as follows:

$$[4] = C_{\lambda'} \circ \mathcal{F} \circ C_{\lambda} \circ \odot^4 \circ \mathcal{F} \circ \odot^4.$$

Conclusion & Open Directions

This model serves as a new baseline. Where do we go from here?

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- **Higher Dimension:** $(4, 4)$ -isogenies
- tbd :)

Thank You!

Questions?